

Syed Maoz Arif

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Electrical engineering undergraduate specialising in PCB design, embedded systems, and IoT. Internationally recognised through the **Altium Global Scholarship (2026)** and **Instructables Sensors Contest Runner-Up (2026)**. Seeking to apply proven hardware design expertise in a professional engineering environment.

EDUCATION

B.Sc. Electrical (Electronics) Engineering — *COMSATS University Islamabad, Abbot-
tabad Campus* **Expected: 2028**

- 4th Semester; coursework: Digital Logic Design, Embedded Systems, Circuit Theory, Signals & Systems.
- Recipient of the **Altium Global Scholarship Program for Emerging Engineers (2026)** — awarded internationally to outstanding hardware design students.

TECHNICAL SKILLS

PCB Design	Altium Designer, Altium 365 (multi-layer routing, schematic capture, DRC, Gerber output)
Embedded / IoT	ESP32, Arduino Nano, STM32, MPU6050; C/C++ firmware, Firebase real-time data streaming
CAD / 3D Modelling	Autodesk Fusion 360, Inventor (parametric modelling, assembly design, 3D-print preparation)
Programming	Python (OOP, automation, simulation), C/C++, $\text{\LaTeX}$ / Overleaf

PROJECTS & RECOGNITION

Omni-Sense Node | *ESP32 · Altium Designer · Firebase · C++*

- Engineered a centralised Data Acquisition and Control System on a single ESP32, integrating **15 peripheral devices** with real-time telemetry streamed to Firebase.
- Designed and fabricated a multi-layer PCB in Altium Designer; implemented sensor polling, data fusion, and remote monitoring firmware.

IncliSense | *ESP32 · MPU6050 · Altium Designer · C++* — **Runner-Up — Instructables Sensors Contest 2026**

- Built a high-precision digital inclinometer with sub-degree angular resolution using the MPU6050 IMU and real-time tilt-compensation algorithms.
- Delivered complete hardware package — custom PCB, enclosure model, and documentation — recognised internationally on Instructables.

EXPERIENCE & LEADERSHIP

Freelance Design Engineer — *Fiverr & Upwork* **2024 – Present**

- Deliver end-to-end PCB design for international clients: schematic capture, multi-layer routing, DFM review, and Gerber file delivery.
- Produce parametric 3D product models and manufacturing-ready assemblies in Fusion 360 and Inventor.

Membership Head — *IET On Campus, Abbottabad* **2024 – Present**

- Directed student recruitment and technical seminars, expanding the chapter's active membership and industry engagement.

Cabinet Member — *Youth Parliament Pakistan* **2025 – 2027**

- Represent KPK in national policy simulation sessions; drafted resolutions on technology, education, and youth development.